Usually we make wave functions by starting with something classical and quantizing it, which gives us an immediate handle on what the wave function is supposed to represent, the basic parts from which it is constructed. But here we don't have any such luxury. All we have is the state itself and Schrödinger's equation. We speak abstractly of "degrees of freedom," but they aren't the quantized version of any readily identifiable classical stuff—they are the quantum-mechanical essence out of which spacetime, and everything else, emerges. John Wheeler used to talk about the idea of "It from Bit," suggesting that the physical world arose (somehow) out of information. These days, when entanglement of quantum degrees of freedom is the main focus, we like to talk about "It from Qubit."

If we look back at the Schrödinger equation, it says that the rate at which the wave function changes with time is governed by the Hamiltonian. Remember that the Hamiltonian is a way of describing how much energy the system contains, and it's a compact way of capturing all of the system's dynamics. A standard feature of Hamiltonians in the real world is dynamical locality—subsystems interact with other subsystems only when they are next to each other, not when they are far away. Influences can travel through space, but only at speeds less than or equal to the speed of light. So an event at one particular moment only immediately affects what's going on at its present location.

With the problem we've assigned to ourselves—how does space emerge from an abstract quantum wave function?—we don't have the convenience of starting with individual parts and asking how they interact. We know what "time" means in this context—it's right there in the Schrödinger equation, the letter $t$—but we don't have particles, or fields, or even locations in a three-dimensional world. We're caught breathing in empty space, and need to look for oxygen where we can find it.

Happily, this is a case where reverse engineering works quite well. Rather than starting with individual pieces of a system and asking how they interact, we can go the other way around: Given the system as a whole (the abstract quantum wave function) and its Hamiltonian, is there a sensible way to break it up into subsystems? It's like buying sliced bread all your life, and then being handed an un-sliced loaf. There are many ways we could imagine slicing it; is there one particular way that's clearly the best?

Yes, there is, if we believe that locality is an important feature of the real world. We can tackle the problem bit by bit, or qubit by qubit, at any